

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS:4

Total Marks:40

1. Develop a system using UML for implementing the Book Bank process. It should verify the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of the student and provide approval to the office. Then the books that are needed by the student will issue from the office to him.
2. Develop a system using UML for implementing the Exam Registration Portal. It should verify the details of the candidate by the central computer. The details regarding the candidate will be provided to the central computer through the administrator and the computer will verify the details of the candidate and provide approval. Then the hall ticket will be issued from the office to the candidate.
3. Develop a system using UML for implementing Stock Maintenance System. In this system, the customer can place orders and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer. Then the stock details are to be updated accordingly.
4. Develop a system using UML for implementing Online Course Reservation System. This system is organized by the central management system. The student first browses and selects the desired course of their choice. The university then checks the availability of the seat if it is available the student is enrolled for the course.

Code of Conduct:

I E. Sakthi priya (192520041) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E. Sakthi priya

RUBRICS FOR CALCULATION:

Experiments

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1) Book Bank Management System

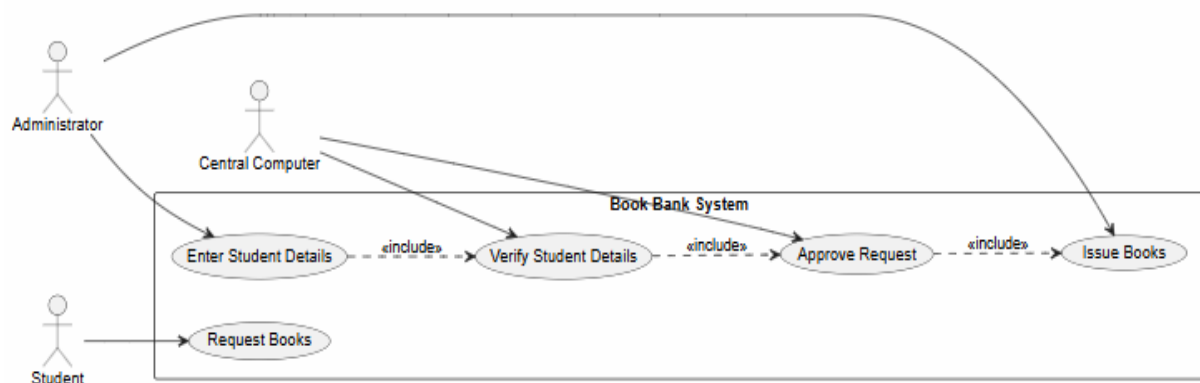
1. Actors

- Student – Requests the required books and provides student details.
- Administrator – Enters student details, checks approval, and issues books.
- Central Computer – Verifies student details and provides approval or rejection.

2. Use Case Diagram

The main use cases are:

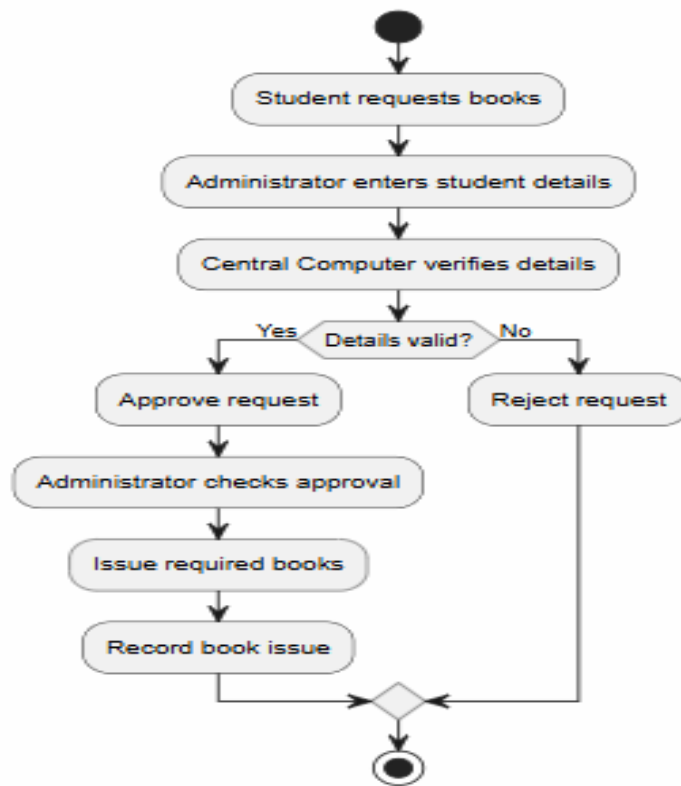
- Request Books
- Enter Student Details
- Verify Student Details
- Approve/Reject Request
- Issue Books



3. Activity Diagram

The Book Bank process follows these steps:

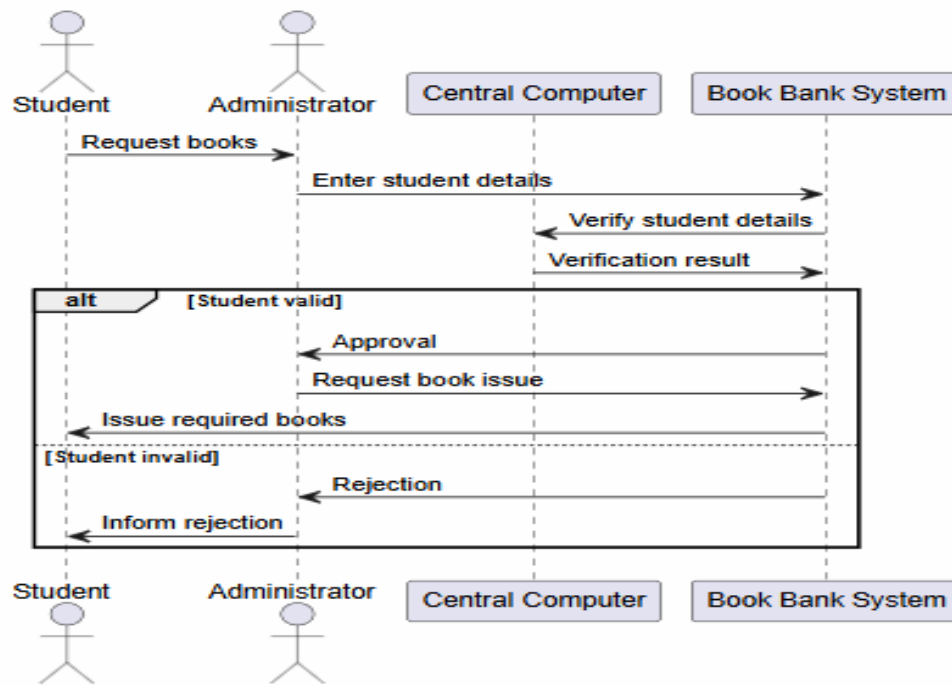
1. Student requests the required books.
2. Administrator enters the student's details.
3. Central Computer verifies the student details.
4. If the details are valid, the request is approved.
5. Administrator checks the approval.
6. Required books are issued to the student.
7. If the details are invalid, the request is rejected.



4. Sequence Diagram

The sequence of interactions is:

1. Student requests books from the Administrator.
2. Administrator enters the student's details into the system.
3. The system sends the details to the Central Computer.
4. Central Computer verifies the student details.
5. The system receives approval or rejection.
6. If approved, the Administrator issues the required books to the Student.
7. If rejected, the student is informed that the request cannot be processed.



5. System Workflow

Student → Administrator → Central Computer → Verification → Approval → Book Issue

2) Exam Registration Portal

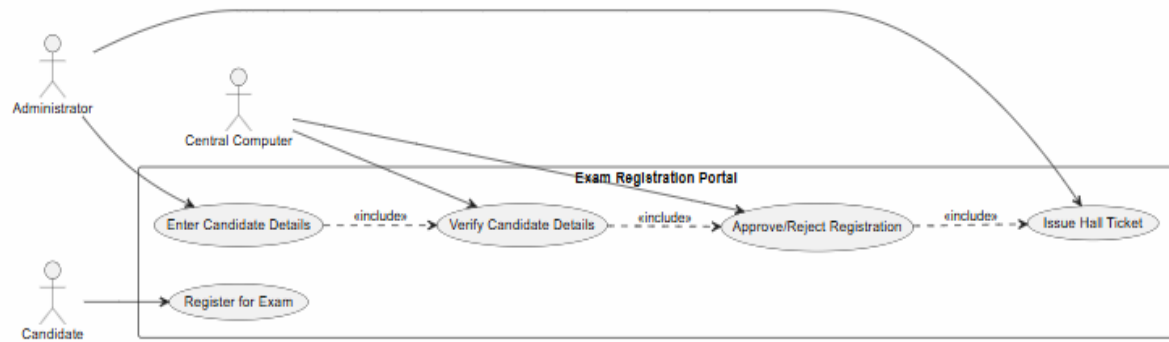
1. Actors

- Candidate – Provides details and registers for the examination.
- Administrator – Enters candidate details and issues the hall ticket.
- Central Computer – Verifies candidate details and provides approval or rejection.

2. Use Case Diagram

The main use cases are:

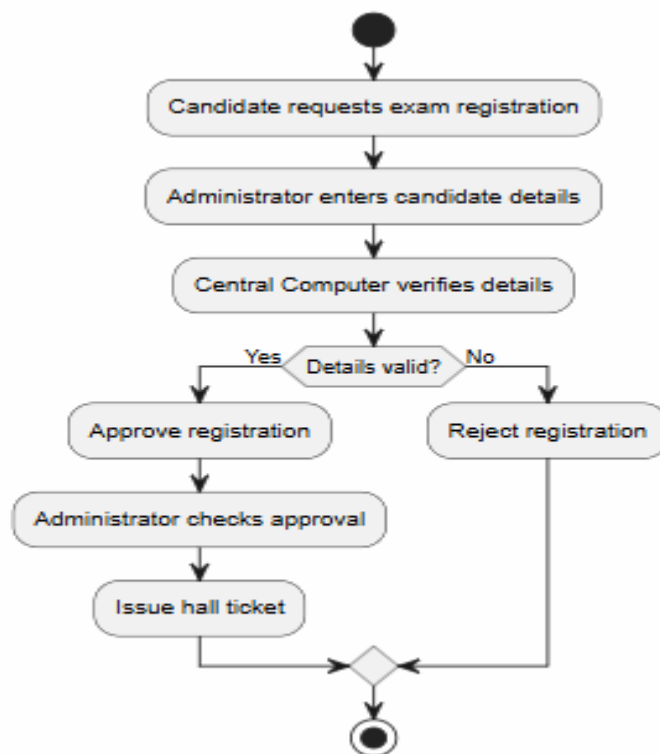
- Register for Exam
- Enter Candidate Details
- Verify Candidate Details
- Approve/Reject Registration
- Issue Hall Ticket



3. Activity Diagram

The Exam Registration process follows these steps:

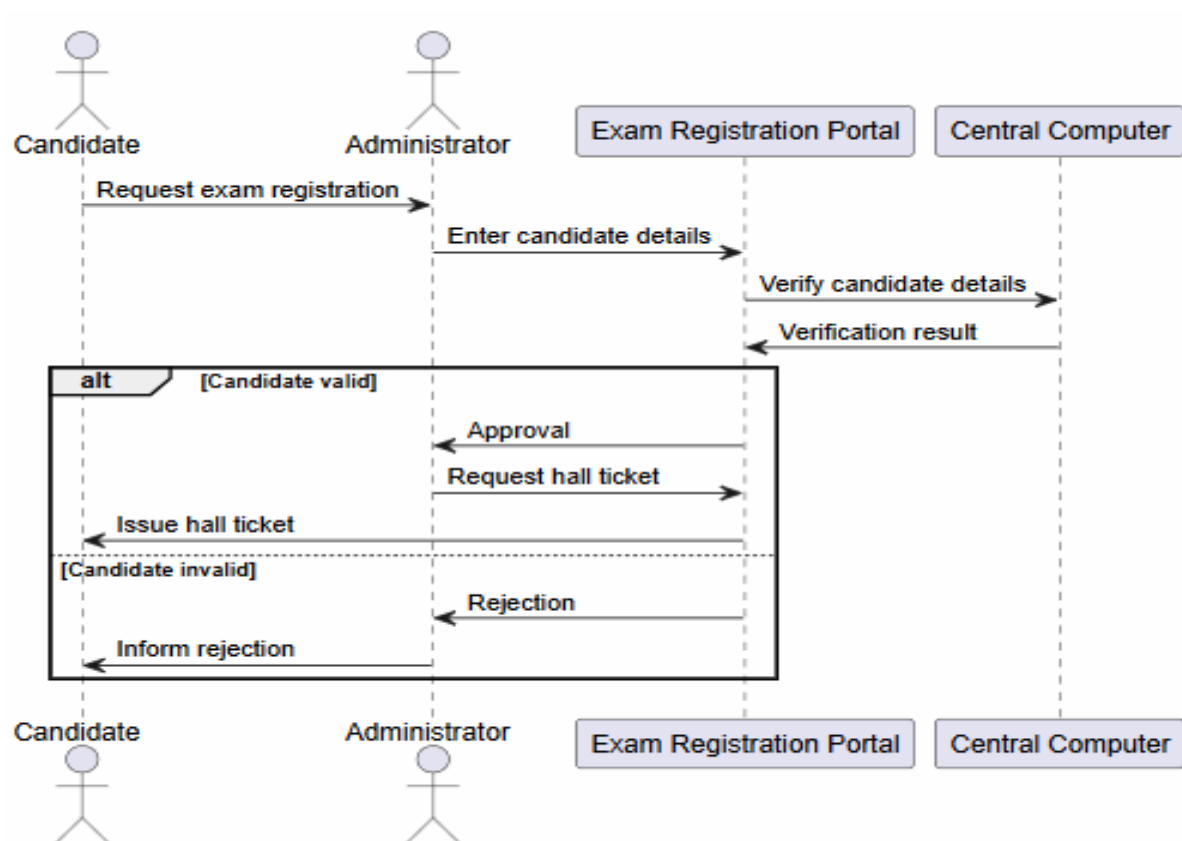
1. Candidate requests exam registration.
2. Administrator enters the candidate's details.
3. Central Computer verifies the candidate details.
4. If the details are valid, the registration is approved.
5. Administrator checks the approval.
6. Hall ticket is issued to the candidate.
7. If the details are invalid, the registration is rejected.



4. Sequence Diagram

The sequence of interactions is:

1. Candidate requests exam registration from the Administrator.
2. Administrator enters the candidate's details into the portal.
3. The portal sends the details to the Central Computer.
4. Central Computer verifies the candidate details.
5. The portal receives approval or rejection.
6. If approved, the Administrator issues the hall ticket to the Candidate.
7. If rejected, the Candidate is informed that the registration is unsuccessful.



3) Stock Maintenance System

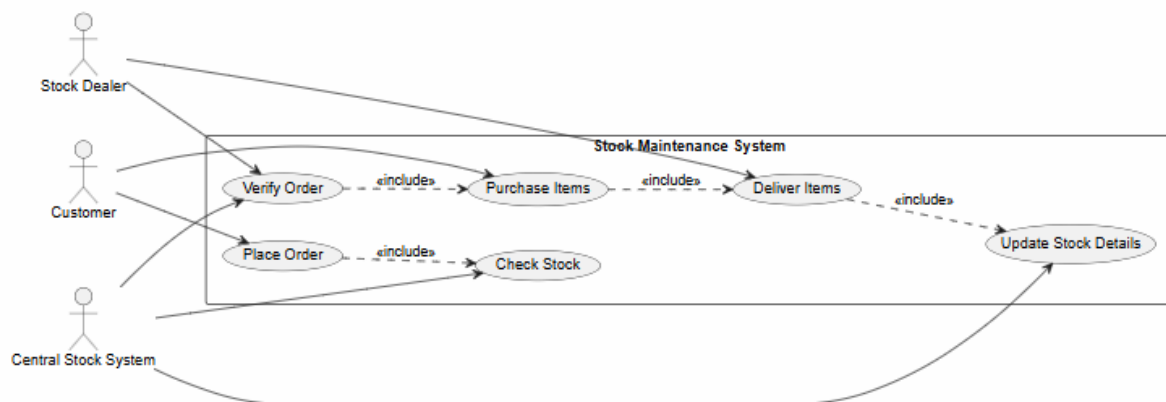
1. Actors

- Customer – Places orders and purchases required items.
- Stock Dealer – Assists the customer, verifies orders, and arranges delivery.
- Central Stock System – Checks and updates stock details and verifies orders.

2. Use Case Diagram

The main use cases are:

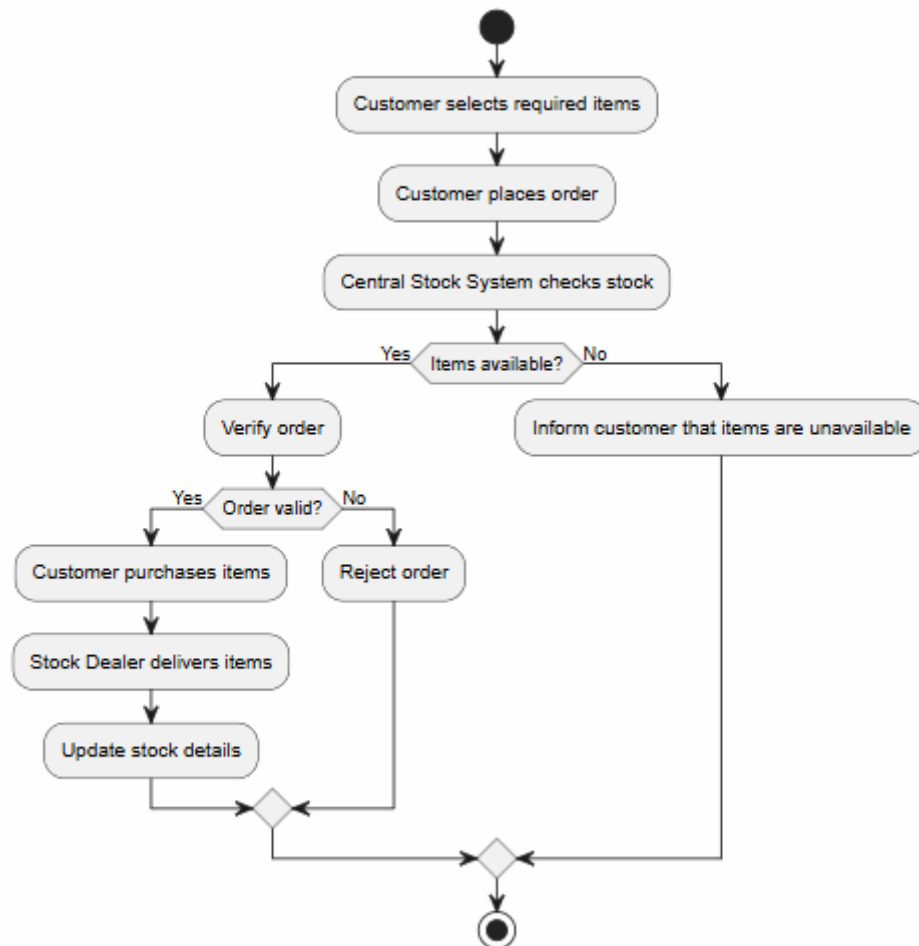
- Place Order
- Check Stock
- Verify Order
- Purchase Items
- Deliver Items
- Update Stock Details



3. Activity Diagram

The Stock Maintenance process follows these steps:

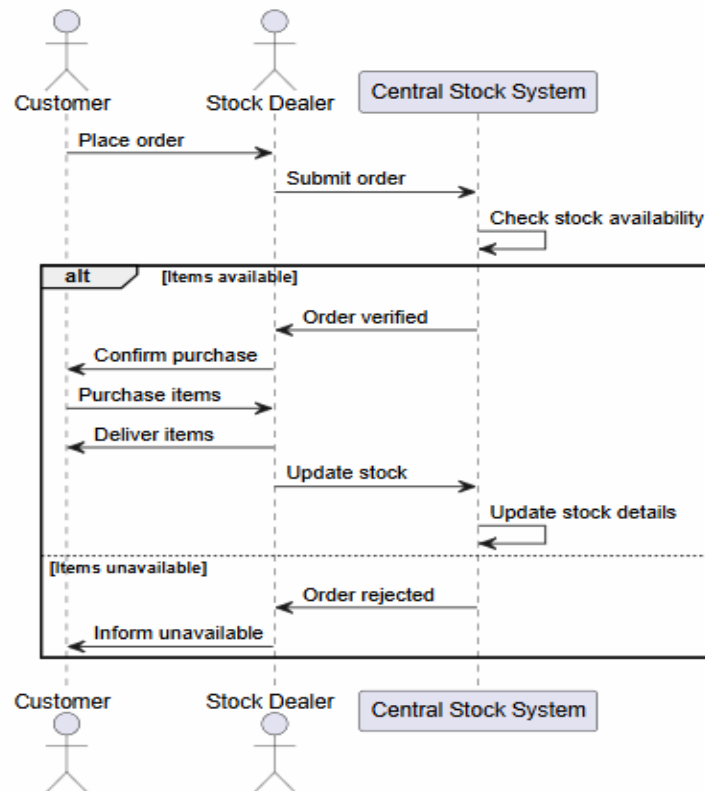
1. Customer selects the required items.
2. Customer places an order with the Stock Dealer.
3. Central Stock System checks the availability of the items.
4. The order is verified.
5. If the order is valid and items are available, the customer purchases the items.
6. The Stock Dealer delivers the items to the customer.
7. The Central Stock System updates the stock details.
8. If the items are unavailable or the order is invalid, the order is rejected.



4. Sequence Diagram

The sequence of interactions is:

1. Customer places an order with the Stock Dealer.
2. Stock Dealer sends the order to the Central Stock System.
3. Central Stock System checks item availability.
4. The order is verified.
5. If approved, the customer purchases the items.
6. Stock Dealer delivers the items to the customer.
7. Central Stock System updates the stock details.
8. If the order is rejected or items are unavailable, the customer is informed.



5. System Workflow

Customer selects items → Places order → Stock System checks availability → Order verified → Customer purchases items → Dealer delivers items → Stock details updated.

4) Online Course Reservation System

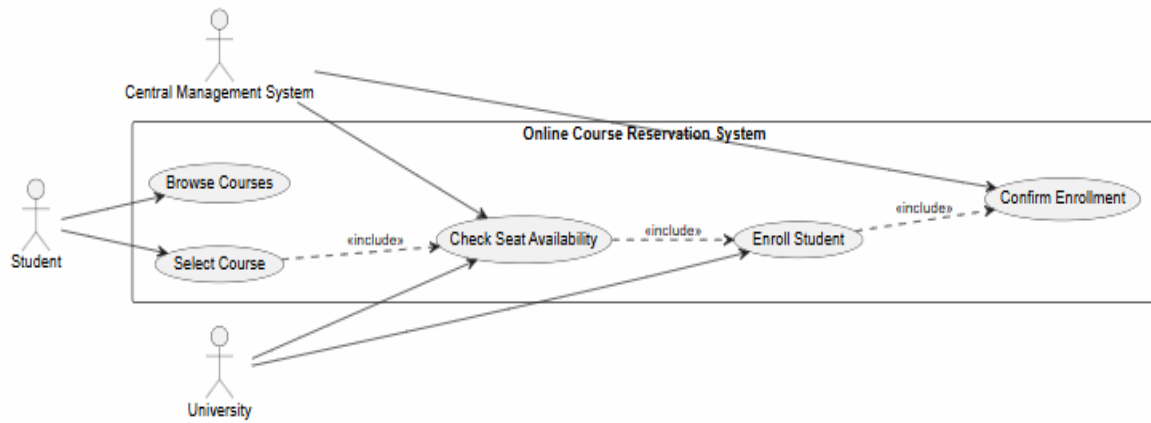
1. Actors

- Student – Browses available courses, selects a desired course, and enrolls in it.
- Central Management System – Manages course information and checks seat availability.
- University – Verifies seat availability and enrolls the student in the selected course.

2. Use Case Diagram

The main use cases are:

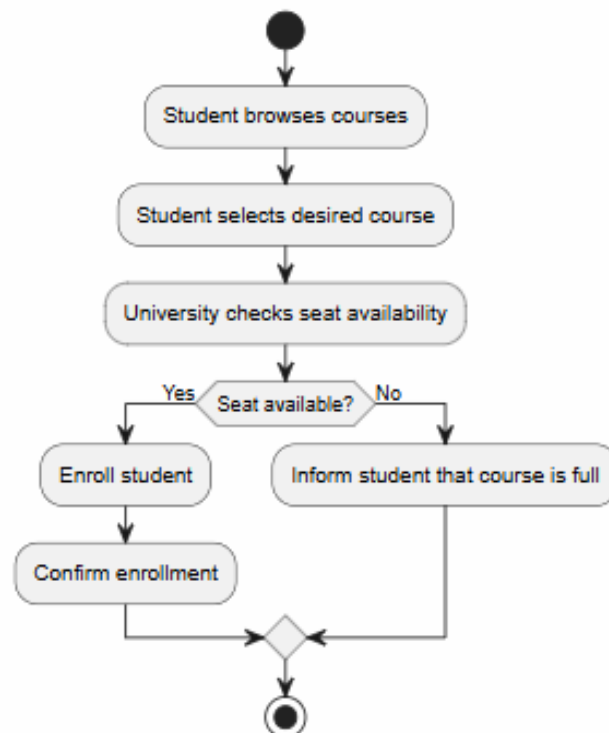
- Browse Courses
- Select Course
- Check Seat Availability
- Enroll Student
- Confirm Enrollment



3. Activity Diagram

The Online Course Reservation process follows these steps:

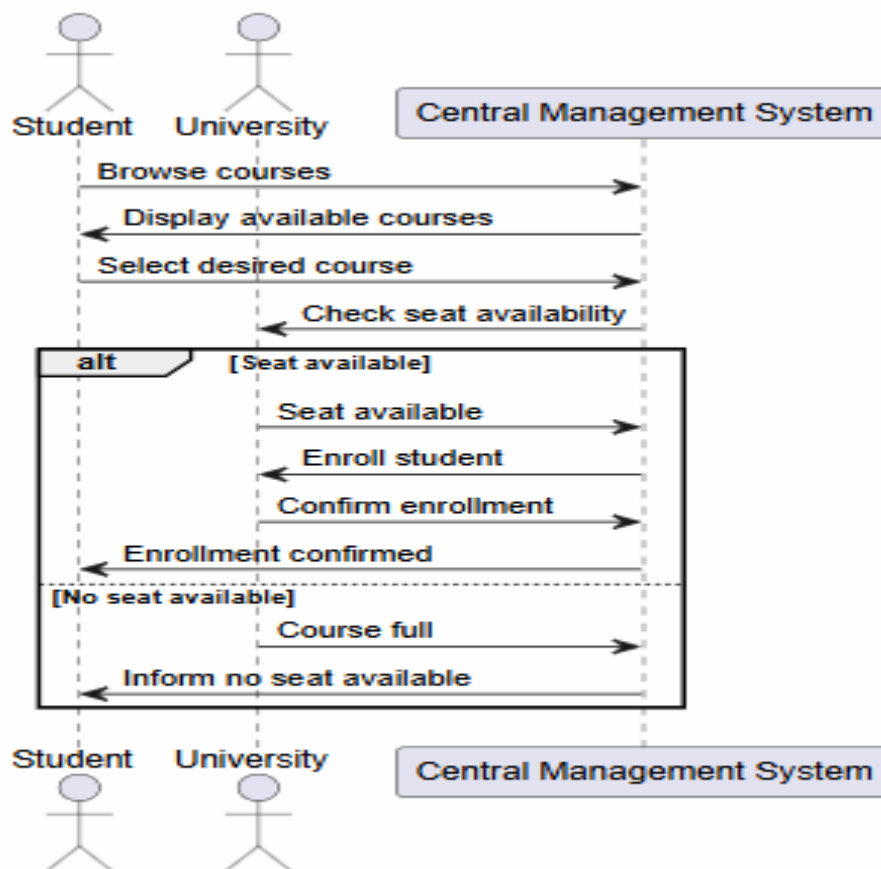
1. Student browses the available courses.
2. Student selects the desired course.
3. University checks the availability of seats.
4. If a seat is available, the student is enrolled in the course.
5. The enrollment is confirmed.
6. If no seat is available, the student is informed that the course is full.



4. Sequence Diagram

The sequence of interactions is:

1. Student browses the available courses.
2. Student selects the desired course.
3. The request is sent to the Central Management System.
4. University checks the availability of seats.
5. If a seat is available, the student is enrolled.
6. Enrollment is confirmed to the student.
7. If no seat is available, the student is informed that the course is full.



5. System Workflow

Student browses courses → Selects desired course → University checks seat availability →
Seat available? → Student enrolled → Enrollment confirmed.
If no seat is available → Student is informed that the course is full.